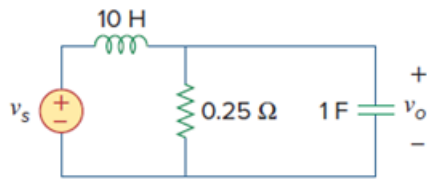


Problem

Find the transfer function V_o/V_s of the circuit in Fig. 14.86. Show that the circuit is a low-pass filter.

Figure 14.86:



Step-by-step solution

Step 1 of 4

Refer to Figure 14.86 in the textbook.

In frequency domain, the inductor is represented as sL , capacitor is represented as $\frac{1}{sC}$.

Draw the circuit diagram in frequency domain.

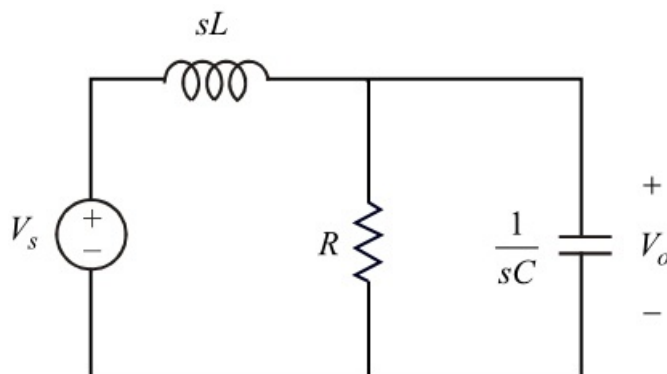


Figure 1

Step 2 of 4

From Figure 1, the output voltage is,

$$\begin{aligned}
 V_o &= V_s \times \frac{R \parallel \frac{1}{sC}}{sL + \left(R \parallel \frac{1}{sC}\right)} \\
 \frac{V_o}{V_s} &= \frac{\left(\frac{R \times \frac{1}{sC}}{R + \frac{1}{sC}}\right)}{sL + \left(\frac{R \times \frac{1}{sC}}{R + \frac{1}{sC}}\right)} \\
 &= \frac{\frac{R}{sC}}{sL \left(R + \frac{1}{sC}\right) + \frac{R}{sC}} \\
 &= \frac{\frac{R}{sC}}{sL \left(\frac{RCs + 1}{sC}\right) + \frac{R}{sC}} \\
 &= \frac{R}{sL(RCs + 1) + R} \\
 \frac{V_o}{V_s} &= \frac{R}{RLCs^2 + Ls + R} \dots\dots (1)
 \end{aligned}$$

Step 3 of 4

Substitute 10 H for L , $0.25\ \Omega$ for R and 1 F for C in equation.

$$\begin{aligned}
 \frac{V_o}{V_s} &= \frac{0.25}{(0.25)(10)(1)s^2 + 10s + 0.25} \\
 &= \frac{0.25}{2.5s^2 + 10s + 0.25} \\
 &= \frac{0.1}{s^2 + 4s + 0.1}
 \end{aligned}$$

Therefore, the transfer function, $\frac{V_o}{V_s}$ is $\boxed{\frac{0.1}{s^2 + 4s + 0.1}}$.

Comments (2)



anonymous1234567jk

why did the resistance become 0.1



Anonymous

they divided everything by 0.25

Step 4 of 4

Substitute $j\omega$ for s .

$$\frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{0.1}{(j\omega)^2 + 4(j\omega) + 0.1}$$

$$H(j\omega) = \frac{0.1}{0.1 - \omega^2 + j4\omega}$$

Calculate the magnitude.

$$|H(j\omega)| = \frac{0.1}{\sqrt{(0.1 - \omega^2)^2 + 16\omega^2}}$$

Calculate the magnitude at $\omega = 0$.

$$\begin{aligned} |H(0)| &= \frac{0.1}{\sqrt{(0.1 - 0^2)^2 + 16(0)^2}} \\ &= 1 \end{aligned}$$

Calculate the magnitude at $\omega = \infty$.

$$\begin{aligned} |H(\infty)| &= \frac{0.1}{\sqrt{(0.1 - \infty^2)^2 + 16(\infty)^2}} \\ &= \frac{0.1}{\infty} \\ &= 0 \end{aligned}$$

From the calculations, it is clear that, the filter circuit allows only low frequencies and attenuates high frequencies.

Therefore, the circuit is low pass filter.